

## Module 2: Assignment

### Data flow and process flow analysis

MCY 630 Security Architecture

#### Assignment Submission Instructions

**Submission requirements:** assignment

**File format:** MSWord or pdf (preferred)

**Report requirements**

- Include your name and assignment title at the top of the document.
- Number the questions and answers and answer each question in your submission in sequence.
- For each response, include the required text as well as any necessary images and diagrams.
- Use black font color for texts
- If uploading an MSWord format file, view the uploaded file from Canvas to make sure that the layout of images and texts are not overlapping in the report. If the contents seem to have moved around, upload a pdf instead of an MSWord file.

#### Instructions – Flow Mapping Tool

In this assignment, you will be utilizing a free-to-use online flow mapping tool to create data flow and process flow maps.

- Draw.IO Online Tool for Flow Map Diagrams: <https://app.diagrams.net/>
- Use the above URL to get to the online interface for your workspace.
- Drag and drop options for various entities are available on the right-hand panel.
- Two objects in the diagram can be connected using connector arrows, which can be dragged from the right-hand panel.
- Alternatively, a faster (and more efficient way) to add entities is to hover your mouse over an object (e.g. a rectangle box). You will notice four arrows appearing as blue shadows. You may choose any of the blue shadow arrows
  - Dragging the mouse on a blue shadow arrow will create a connector arrow for that object, which can then be connected by releasing the mouse on top of another object.
  - Clicking the mouse on a blue shadow arrow will display the most common flow objects, any of which can then be selected by another mouse click, and the next object will automatically be added and connected to the present object.
- Keep in mind that you may need to adjust the placement of the connectors and objects for improving the layout of your flow maps based on the connected edges and cross-over connectors.
- You may add labels to any objects (entities and connectors) by selecting the item and just typing in the text.
- Explore the options for adding properties for objects (right click -> Edit Data).
- Explore the top-panel buttons (File, Edit, View, etc.) for further features (e.g. saving draw.io XML files and importing them later, exporting created flow maps as images, etc.)

## Scenario

*Consider the following system application scenario*

A simple online food pick-up ordering application platform for a restaurant, MCY Fast Food, allows online users to perform the following actions using the web-based application interface:

- Browse through the restaurant's menu options, corresponding prices, and nutrition information.
- Add items using a "add" button to the cart.
- View the cart to finalize the order.
- Items can be deleted from the cart before proceeding to finalize the order.
- Provides all necessary personal user information and pay the total amount due using a credit card gateway.
- Receive confirmation email with order details, itemized bill, and approximate pick up time.
- The restaurant also receives an email with the order details, itemized bill, and approximate pick up time for the customer.

## Questions

Answer the following questions considering the limited view of the application presented above.

Do note that it is acceptable if you have assumptions. However, you will need to state your assumptions (e.g. the pick-up time is automatically generated by the system using pre-configured times required to prepare each item) and the justifications behind the assumptions (e.g. the restaurant chef already knows how much time is needed to prepare an item). Assumptions may sometimes help you avoid unnecessary complications. However, be careful to not exclude important aspects of the system by adopting assumptions for the sake of simplicity.

1. (10 points) Identify the possible data items which are exchanged during the use of the online ordering application for MCY Fast Food, the type of each data item, and the compliance standard (if applicable).
2. (20 points) Draw the data flow map for the online ordering application for MCY Fast Food. Ensure that your entities and connectors are all properly labelled. Export the map as an image to include it in your submission document. Include a short block of text to explain the data flow map as seen from your justifications.
3. (20 points) Draw the process flow map for the online ordering application for MCY Fast Food. Ensure that your entities and connectors are all properly labelled. Export the map as an image to include it in your submission document. Include a short block of text to explain the process flow map as seen from your justifications.